WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Asking Questions with Data

Problems, questions and hypotheses



What problems can data science answer?

Type	Ask it like this	Everyday example
Describe	What happened? How much?	What is the average order size at the campus store?
Compare	Is group A different from group B?	Do evening classes have more absences than morning classes?
Relate	Do these two things move together?	Do students who study longer score higher?
Predict	What will happen next?	How many people will visit the library next Monday?
Group	Are there natural groups?	What kinds of listeners use a music app?

What data cannot answer alone

Taste and opinion

"Is jazz better than pop?"

Better: Which genre had more streams last month?

Proof of cause

"Does the discount cause more returns?"

Better: Do campaigns with bigger discounts have more returns?

Right and wrong

"Should we track student phones?"

Better: What data would we collect, and who would it affect?

What makes a good question?

Specific	Says who, what and when.
Measurable	A column in a dataset could answer it.
Answerable	We can actually get the data.
Useful	Someone will act on the answer.
Responsible	Collecting the data is fair and legal.

From weak to strong questions

Weak question	What is wrong	Better question
Is pizza good?	Opinion. Nothing to measure.	Which pizza place delivers fastest on Fridays?
Do students like the library?	Too vague. Who? When?	How many students visit the library each day during finals week?
Why do people buy things?	Too big to answer.	Do customers who get a discount spend less per order?

From problem to hypothesis

1 Problem statement

The situation and why it matters.

Example:

The café throws away muffins every day.

2 Questions

What we want to find out. One idea each.

Example:

Do rainy days change muffin sales?

3 Hypothesis

Our testable guess, with a reason.

Example:

Rainy days sell more muffins.

4 Check with data

Does the data agree, or not?

Example:

Compare rainy days with dry days.

What is a hypothesis?

A testable guess about how two things are related.

Testable: Data can check it.

Specific: Names the columns and the direction.

Can be wrong: If nothing could prove it wrong, it is not a hypothesis.

Weak

"Sales will change."

Strong

"Campaigns with bigger discounts have a higher share of returned orders."

The hypothesis recipe

I think ____ (X) is related to ____ (Y), because ____ (reason).

Example: campus café

The hypothesis

I think rainy days have higher muffin sales than dry days, because people stay inside and buy a snack with coffee.

How we check it

Supported if rainy-day sales are higher.

Not supported if they are the same or lower.

Columns: Weather, Muffins_Sold

Dataset 1: Marketing campaigns

An online shop ran 10 marketing campaigns. Each row is one campaign. (Made-up data)

Campaign ID	Channel	Month	Budget USD	Discount Pct	Audience Size	Clicks	Orders	Revenue USD	Returned Orders
C01	Email	Jan	500	10	20000	1800	270	12150	14
C02	Social	Feb	1500	20	50000	2500	160	5280	16
C03	Search Ads	Mar	2000	10	30000	3000	330	14850	20
C04	Flyer	Apr	800	15	8000	240	36	1440	4
C05	Email	May	600	20	22000	2400	390	12870	47
C06	Social	Jun	1200	10	45000	1800	84	3780	5
C07	Search Ads	Jul	2500	25	35000	4200	610	18300	91
C08	Flyer	Aug	700	30	7500	300	66	1782	13
C09	Email	Sep	400	15	18000	1500	205	8200	20
C10	Social	Oct	1800	30	52000	3100	248	6696	40

Dataset 2: Novels, 1818 to 1897

A library counted features in 10 made-up novels. Each row is one book.

Book ID	Year	Genre	Narration	Words k	Chapters	Avg Sentence Len	Fear Words per 1000	Female Char Pct	Editions
B01	1818	Gothic	Third	142	31	29	8.5	22	3
B02	1826	Romance	First	118	27	27	1.1	41	5
B03	1834	Gothic	First	96	22	28	7.9	30	2
B04	1845	Adventure	Third	210	48	24	2.2	12	9
B05	1853	Mystery	Third	155	36	23	4.6	28	7
B06	1861	Romance	Third	175	40	25	1.4	46	6
B07	1872	Gothic	Third	128	29	21	9.3	35	4
B08	1880	Adventure	First	188	42	20	2.8	18	11
B09	1889	Mystery	Third	139	33	19	5.1	38	8
B10	1897	Romance	First	110	25	18	1.0	52	7